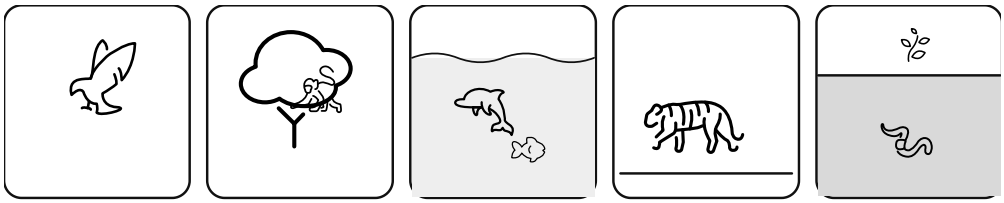
LIVING THINGS

Animals of Bangladesh: Diversity and Classification

Where animals live, vertebrates and invertebrates, skeletons, keys, adaptation and migration



aerial *in the air* | **arboreal** *in trees* | **aquatic** *in water* | **terrestrial** *on land* | **fossorial** *under the ground*

In this pack

Learn it: Animals of Bangladesh: so many kinds! ★☆☆
Fact check: Sorting animals ★☆☆
Think it through: Adapted to survive ★★☆
Investigate: Minibeast hunt in the school garden ★★☆
Real-world case study: Winter visitors: the birds of Tanguar Haor ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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Animals of Bangladesh: so many kinds!

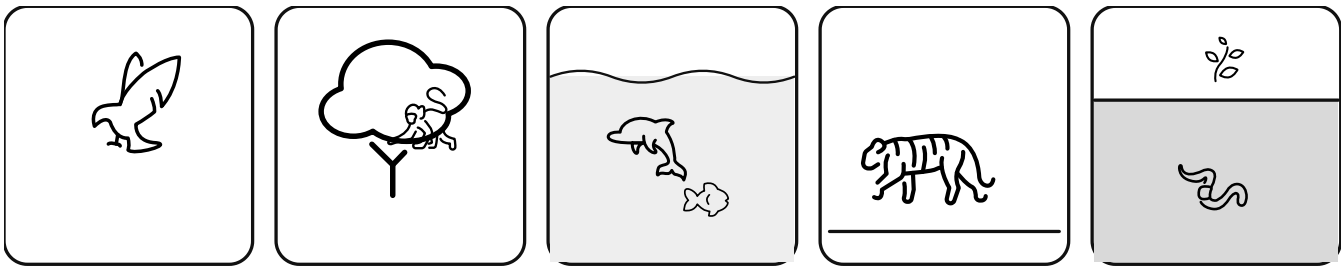
★☆☆ Grade 4

Name: Class: Date:

Learning goals

- Name the places where animals live: in the air, in trees, in water, on land and under the ground.
- Sort animals into groups and use a branching key.
- Explain adaptation and migration with examples from Bangladesh.

Bangladesh is a small country, but it is full of animals. Tigers hide in the Sundarbans, dolphins swim in the rivers, hoolock gibbons swing through the forests of Sylhet, and every winter thousands of ducks fly in from the cold north. Scientists have counted more than 1,600 kinds of animals with backbones here, and many thousands of insects. This huge variety of living things is called **biodiversity**.



aerial *in the air* | **arboreal** *in trees* | **aquatic** *in water* | **terrestrial** *on land* | **fossorial** *under the ground*

Animals live in different places. Each place gives them food, water and shelter.

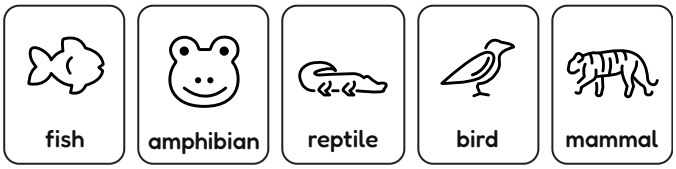
Where do animals live?

- **Aerial** animals spend much of their life flying in the **air**: eagles, swallows, bats.
- **Arboreal** animals live in **trees**: monkeys, gibbons, squirrels, many birds and snakes.
- **Aquatic** animals live in **water**: fish such as the hilsa, Ganges river dolphins, crabs, frogs (for part of their life).
- **Terrestrial** animals live on **land**: tigers, deer, elephants, cows and people.
- **Fossorial** animals live **under the ground**: earthworms, moles, many ants and some snakes.

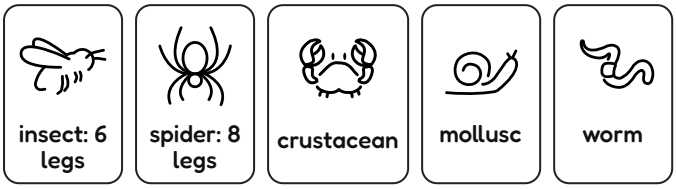
Two big groups

Scientists sort animals into groups by looking at their bodies. The first question is: does it have a **backbone**?

Vertebrates have a backbone and a skeleton **inside** their bodies. There are five groups: **fish**, **amphibians** (frogs, toads), **reptiles** (snakes, lizards, crocodiles, turtles), **birds** and **mammals** (tigers, bats, people).

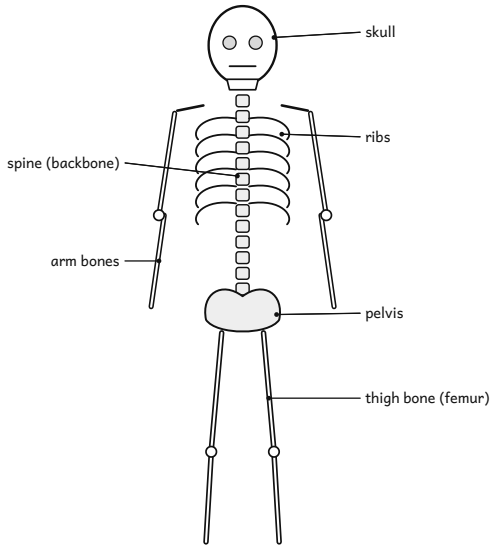


Invertebrates have **no backbone**. Some, like insects, spiders and crabs, have a hard **exoskeleton** on the **outside**, like a suit of armour. Others, like earthworms and jellyfish, are soft. More than nine out of ten kinds of animals are invertebrates!



Skeletons inside us

Your skeleton is a strong frame of more than 200 **bones**. It **supports** your body, **protects** soft parts (your skull protects your brain, your ribs protect your heart and lungs) and helps you **move**. **Muscles** pull on bones to move them. Muscles can only pull, so they work in pairs: one pulls the bone one way, the other pulls it back.



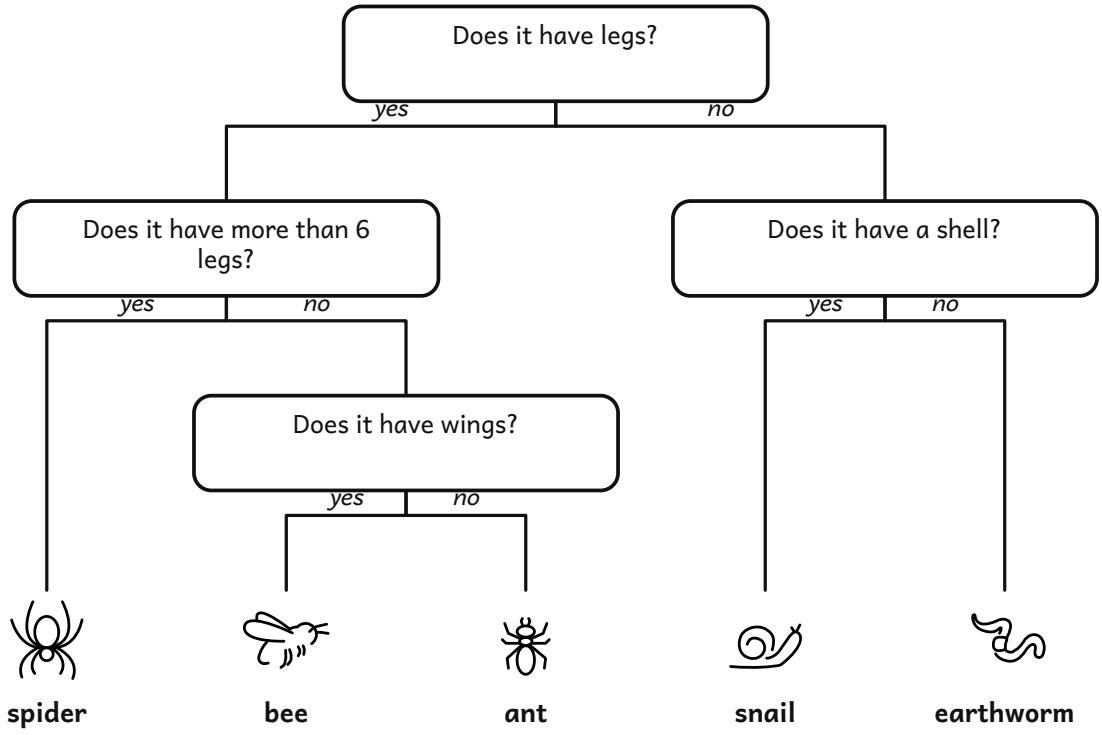
The human skeleton.

Did you know?

Sharks have no bones at all! Their skeleton is made of **cartilage**, the same bendy material as the tip of your nose.

Branching keys

A **key** helps you name a living thing by answering yes/no questions about what you can see. Start at the top and follow the answers down the branches until you reach a name.



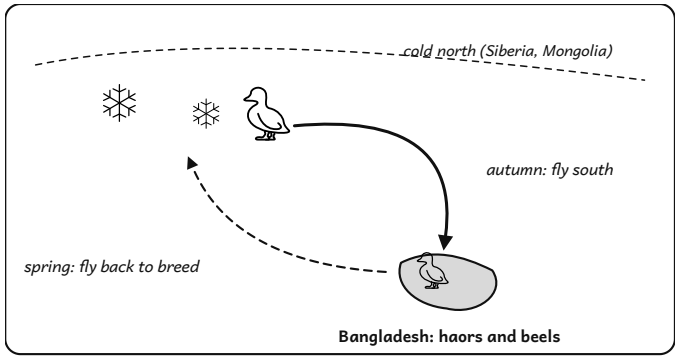
A branching key for some small animals.

Adaptation

An **adaptation** is a feature that helps an animal **survive** in its habitat. The tiger's stripes hide it in tall grass. The Ganges river dolphin is almost blind, so it "sees" with sound (echoes) in the muddy water. Fishing cats have webbed toes for swimming. Birds' beaks are shaped for their food: a kingfisher's long sharp beak catches fish, a parrot's strong curved beak cracks seeds.

Migration

Migration is when animals travel from one place to another, usually at the same time each year, to find food, warmth or a safe place to breed. Every winter, ducks and waders fly thousands of kilometres from cold lands like Siberia to the **haors** and **beels** (wetlands) of Bangladesh. Hilsa fish swim from the sea up our rivers to lay eggs.



Winter visitors.

When habitats change

Natural changes, such as **floods**, **droughts**, **cyclones** and rivers changing course, can change a habitat. Some animals move away or die, while others find new homes. If a habitat changes too fast, a kind of animal may disappear from an area for ever. When the last one dies everywhere, it is **extinct**, like the dodo.

Bangladesh connection

Some animals that once lived in Bangladesh are now gone from here, such as the Indian rhinoceros and the wild water buffalo. The Royal Bengal tiger, the national animal, is protected in the Sundarbans.

Key words

biodiversity	the variety of living things in a place
vertebrate	an animal with a backbone
invertebrate	an animal without a backbone
exoskeleton	a hard covering on the outside of an animal
adaptation	a feature that helps a living thing survive
key	yes/no questions that help you name a living thing
migration	travelling to another place at the same time each year
habitat	the place where a living thing lives

FACT CHECK · A1 · ANIMAL DIVERSITY

Sorting animals

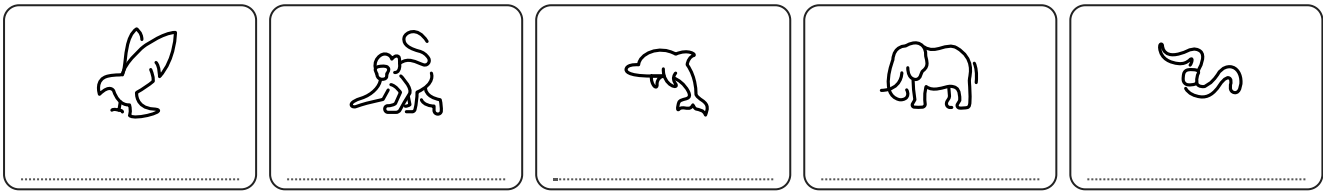
★☆☆ Grade 4

Name: Class: Date:

Learning goals

- Sort animals by where they live.
- Name vertebrate groups.
- Use a key.

1 **Where does each animal live? Write aerial, arboreal, aquatic, terrestrial or fossorial.** [5]



2 **Sort the animals.** [8]

SORT THESE

earthworm kingfisher hilsa fish tiger butterfly frog crab spider

Vertebrate (backbone)	Invertebrate (no backbone)

3 **Match each animal to its vertebrate group.** [5]

Draw a line to match each pair, or write the letter.

crocodile ●	● **A** mammal
toad ●	● **B** bird
bat ●	● **C** fish
owl ●	● **D** amphibian
shark ●	● **E** reptile

4 **Use the key on the Learn it page. Which animal has legs, but not more than 6 legs, and has wings?** [1]

.....

5 **Fill in the gaps.** [4]

WORD BANK

Muscles **migrate** **pushes** **protects** **supports** **eats**

Your skeleton _____ your body. Your skull _____ your brain.

_____ pull on bones to make them move. Animals that travel to another place every year _____.

THINK IT THROUGH · A1 · ANIMAL DIVERSITY

Adapted to survive

★★☆
Grade 4

Name: Class: Date:

Learning goals

- Explain adaptations.
- Make your own key.

1 **Explain how each feature helps the animal survive.** [4]

Animal and feature	How it helps
tiger: striped coat	
river dolphin: uses echoes	
duck: webbed feet	
camel: wide feet	

2 **Look at the birds' beaks. Match each beak to its food: fish, seeds and nuts, nectar in flowers, meat.** [4]

Beak	Food
long and sharp like a spear (kingfisher)	
short, strong and curved (parrot)	
long and thin like a straw (sunbird)	
sharp and hooked (eagle)	

3 **Make a branching key to sort these four animals: fish, bird, snake, cow. Write a yes/no question in each box.** [4]

My key

4 **Why do ducks fly to Bangladesh in winter, and why do they fly back in spring?** [2]

INVESTIGATE · A1 · ANIMAL DIVERSITY

Minibeast hunt in the school garden

★★☆ Grade 4

Name: Class: Date:

Learning goals

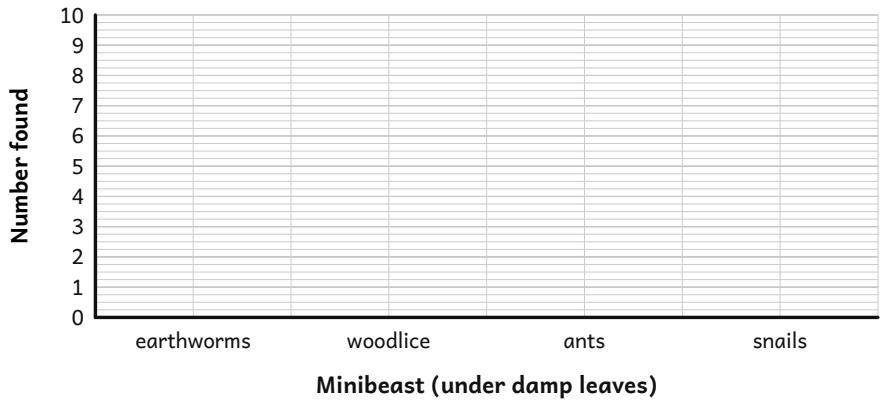
- Record observations in a tally chart.
- Draw a bar chart.
- Make conclusions.

Class 4B searched two places in the school garden for 10 minutes each: under a pile of damp leaves and on a sunny path. They counted the minibeasts (small invertebrates) they found.

1 **Complete the totals.** [2]

Minibeast	Under damp leaves	On the sunny path
earthworms	IIII I	I
woodlice	IIII IIII	0
ants	III	IIII IIII II
snails	IIII	0
Total		

2 **Draw a bar chart for the damp leaves.** [3]



3 **Which place had more kinds of minibeasts? Suggest why.** [2]

.....

.....

4 **Was the test fair? Name one thing they kept the same.** [1]

.....

5 **How should they look *a*fter the minibe*a*sts?** [1]

Winter visitors: the birds of Tanguar Haor

★★★ Grade 4

Name: Class: Date:

Learning goals

- Read about a real wetland.
- Use data about bird numbers.
- Suggest ways to protect migrating birds.

Science · Geography · Maths · Citizenship

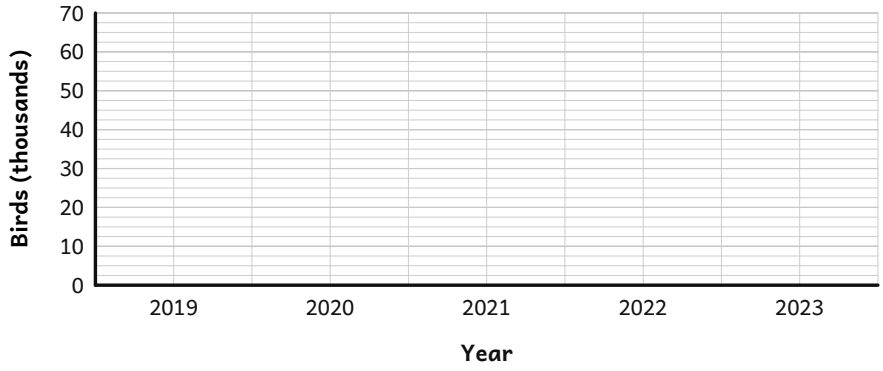
Tanguar Haor in Sunamganj is a huge wetland of lakes, marshes and swamp forest. In winter, when the water goes down, it becomes home to tens of thousands of **migratory birds**: ducks, geese, cranes and waders that fly from Siberia, Mongolia and the Himalayas. Because it is so important, it is a protected **Ramsar site** (a wetland of international importance).

Every year, bird scientists and volunteers count the birds. The counts show how healthy the wetland is. The birds face dangers: some people hunt them or catch them with poison bait, fishing nets can trap them, and too many boats disturb them. Local people, the government and nature groups work together to protect the haor.

Based on information about Tanguar Haor and the Asian Waterbird Census. Counts below are example data.

1 A bird count at a wetland (example data). [3]

Year	2019	2020	2021	2022	2023
Birds counted (thousands)	50	45	60	38	52



a) Draw the bar chart.

b) Which year had the most birds? Which had the fewest?

2 **Why do scientists count the birds every year?** [2]

.....

.....

3 **What could each person do to help the birds?** [3]

Person	Action
a fisher	
a tourist	
a learner like you	

4 **Design a poster to protect winter visitors. Include a drawing and a short message.** [2]

My poster

Show what you know

★★★ Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

D	F	E	H	T	L	G	G	N	A	M
M	I	G	R	A	T	E	M	J	I	I
I	N	I	D	G	B	G	A	F	F	F
N	R	R	J	C	A	R	M	A	A	N
S	A	F	S	C	C	E	M	D	J	H
E	S	G	N	P	K	P	A	A	L	A
C	U	K	O	B	B	T	L	P	W	B
T	I	K	S	M	O	I	I	T	P	I
Y	B	E	P	A	N	L	M	W	E	T
V	N	Y	S	A	E	E	L	P	A	A
M	S	K	E	L	E	T	O	N	W	T

adapt backbone habitat insect key mammal migrate reptile skeleton

2 Which animal is an invertebrate? [1]

- ◯ frog
- ◯ snake
- ◯ crab
- ◯ bat

3 A fossorial animal lives... [1]

- ◯ in trees
- ◯ under the ground
- ◯ in the air
- ◯ in water

4 Write the job of each part of the skeleton. [3]

Part	Job
skull	
ribs	
backbone (spine)	

5 **Challenge. A new road is built through a forest. Explain how this could change the biodiversity of the forest.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
name where animals live: aerial, arboreal, aquatic, terrestrial, fossorial	○	○	○
sort animals into vertebrates and invertebrates	○	○	○
name the five vertebrate groups	○	○	○
use and make a branching key	○	○	○
explain adaptation and migration with examples	○	○	○
explain how we can protect migrating birds	○	○	○